

Ex. No. : 07

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A PYTHON PROGRAM TO IMPLEMENT DECISION TREE

Aim:

To implement a **Decision Tree model in Python**

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('Iris.csv')

X = df[['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm']]
y = df['Species']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))

plt.figure(figsize=(12,8))
```

```
plot_tree(model, feature_names=X.columns, class_names=model.classes_,
filled=True)
plt.title("Decision Tree Structure")
plt.show()
```

```
X_vis = X[['SepalLengthCm','SepalWidthCm']].values
y_vis = y
```

```
model_vis = DecisionTreeClassifier()
model_vis.fit(X_vis, y_vis)
```

```
x_min, x_max = X_vis[:,0].min()-1, X_vis[:,0].max()+1
y_min, y_max = X_vis[:,1].min()-1, X_vis[:,1].max()+1
```

```
xx, yy = np.meshgrid(
    np.linspace(x_min, x_max, 100),
    np.linspace(y_min, y_max, 100)
)
```

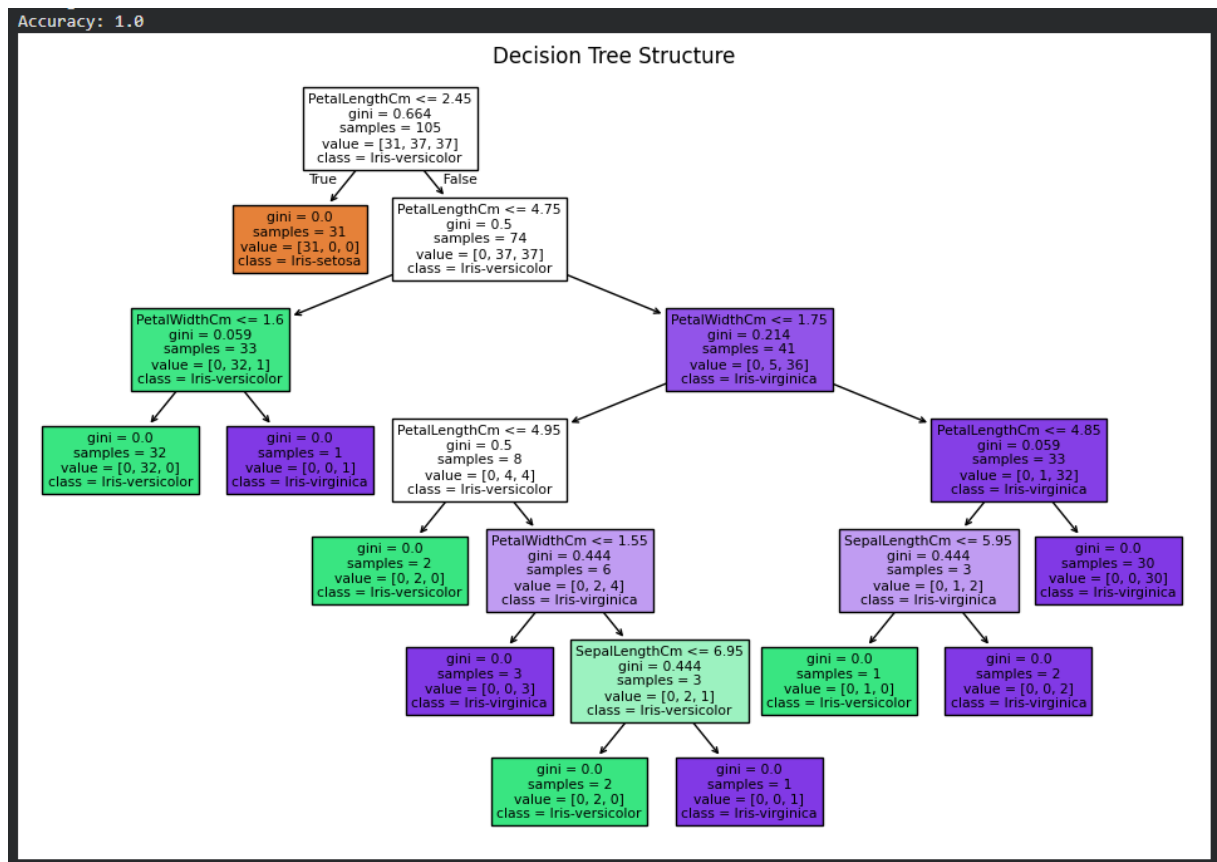
```
Z = model_vis.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
```

```
plt.figure()
plt.contourf(xx, yy, Z)
```

```
for label in np.unique(y_vis):
    plt.scatter(
        X_vis[y_vis == label, 0],
        X_vis[y_vis == label, 1],
        label=label
    )
```

```
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.title("Decision Boundary (Decision Tree)")
plt.legend()
plt.show()
```

OUTPUT:



Result:

Thus, the Decision Tree Classifier was successfully implemented using Python